



COPPERBELT UNIVERSITY
SCHOOL OF MEDICINE
DEPARTMENT OF BASIC SCIENCES
SESSIONAL EXAMINATION 2018/19

BASIC HUMAN PATHOLOGY MBS 220

STUDENT NUMBER	
PROGRAM	

15006072

MBChB

DATE: 04th OCTOBER, 2019

TIME: 09:00 – 12:00 HRS

DURATION: 3 HOUR

149

INSTRUCTIONS TO CANDIDATES

- IN THE BOXES ABOVE, WRITE YOUR FULL NAMES & STUDENT NUMBER CLEARLY
- CHECK THAT YOU HAVE THE CORRECT QUESTION PAPER
- ANSWER ALL THE QUESTION IN THE SPACES & PAPER PROVIDED

INFORMATION TO CANDIDATES

- ANSWER ALL QUESTIONS IN SECTION A, B, C & D
- THE PAPER HAS A TOTAL OF 100 MARKS

SECTION A. MULTIPLE CHOICE QUESTIONS. CHOOSE ONE CORRECT CHOICE.

1. Prognosis include all of the following except
 - A. Complete recovery
 - B. Chronic or partial recovery
 - C. Death
 - ☒ D. Recovery with death
2. Regarding a LESION?
 - ☒ A. When the change is specifically characteristic of a disease is called Pathognomonic lesion
 - B. There is no biochemical changes to the cells
 - C. Is not associated with structural changes
 - D. Is always a gross lesion
3. The causes of fatty change include the following: ^{except}
 - ☒ A. Alcohol abuse
 - B. Protein malnutrition
 - C. Diabetes insipidus
 - ☒ D. Dysplasia
4. Free radicals can be involved in the following excerpt?
 - A. Cause cell injury
 - B. Can initiate autocatalytic reactions
 - C. Are generated during metabolism
 - ☒ D. Are not involved in the disease process
5. Metastatic calcification:
 - A. Is caused by consumption of large quantities of calcium
 - B. Occurs in previously damaged tissues
 - C. Commonly seen in benign tumours
 - ☒ D. The serum calcium level is elevated
6. The following are causes of embolism excerpt
 - ☒ A. Inflammatory cytokines
 - B. Amniotic fluid
 - C. Bits of bone marrow
 - D. Parasites - *Dirofilariasis*
7. Crigler-Najjar syndrome type 1 is a condition characterised by:
 - ☒ A. Complete absence of UGT1A1 enzyme
 - B. Partial deficiency of the UGT1A1 enzyme
 - C. Biliary colic
 - D. Severe headache

8. Which of the following is NOT true about Liver cirrhosis is:
- A. Regeneration process of the liver.
 - B. Scarring of the liver.
 - C. Scarring and regeneration of the liver.
 - ☒ D. None of the above.
9. Regarding aneurysms:
- ☒ A. Atherosclerotic aneurysms are common true aneurysms
 - B. Fluid extravasation out of the vascular system is common
 - C. Is the same as dissections
 - D. Is a localised transient dilatation in the blood vessel
10. Some of the complication of Acute rheumatic fever include the following excerpt:
- A. Mural thrombi and embolism
 - ☒ B. Infective endocarditis due to deformed valves
 - C. CCF
 - D. None of the above
11. Which one of the following statements is true about chronic inflammation?
- A. It has two parts, granulomatous and granulation
 - B. Divided into two parts, granulation and non-granulomatous
 - ☒ C. The key cell is lymphocytes
 - D. The characteristic cell is eosinophil
12. Malignant transformation in gastric ulcer is considered as a....
- A. Clinical sign
 - ☒ B. Clinical complications
 - C. Indications for increased acid production
 - D. The cause of cardio spasm
13. All the following are risk factors for the tumors of the stomach **except**
- A. Cigarette smoking
 - B. Gastric Peptic ulcer
 - ☒ C. Dyspepsia
 - ☒ D. Dysplasia
14. The following statements is correct about pyloric stenosis
- A. More commonly seen in female children
 - ☒ B. There is projectile, non-bilious vomiting in 4th or 5th week of life
 - C. There is visible peristalsis, firm ovoid palpable mass in region of upper stomach
 - ☒ D. May be Secondary to hypertrophy and hyperplasia of muscularis of pylorus
15. About Autoimmune gastritis
- ☒ A. May lead to megaloblastic anaemia
 - B. May results in increased secretions of intrinsic factor
 - ☒ C. May cause iron deficiency anaemia
 - D. May lead to lack of intrinsic factor hence causing hemorrhagic anaemia

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16. Which one of the following is correct about aphthous ulcer?
- A. Painful, superficial ulceration of stomach
 - B. May arise due to HIV infections.
 - ☒ C. May arise in relation to stress and resolves spontaneously, but often recurs
 - D. May arise due to stress but never recur
17. Sialadenitis.....
- A. Always follows malignant lesions
 - B. Results from Tuberculosis infections
 - ☒ C. Staphylococcus aureus is mostly involved
 - ☒ D. Mostly affects the Submandibular gland
18. Pleomorphic adenoma.....
- A. Fixed and painful mass
 - B. Most common in the submandibular
 - ☒ C. The benign tumour of the parotid gland
 - ☒ D. The malignant tumour of the salivary gland
19. Which of the following best describe cleft lip and palate
- A. Full thickness defect of head or neck
 - ☒ B. Defect due to failure of facial prominences to fuse
 - C. Cleft and palate usually occur together among black populations
 - D. Full defect which involves palate and tongue
20. Which one of the following tumours described as benign cystic tumour with abundant lymphocyte and germinal centres?
- A. Pleomorphic adenoma
 - ☒ B. Warthin tumour
 - C. Mucoepidermoid tumour
 - D. Malignant tumour of parotid

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SECTION B. INDICATE TRUE OR FALSE ON EACH GIVEN OPTION

1. Analyse the following statements and decide whether they are true or false:-

- ~~F~~ A. Hernia an abnormal or surgically made passage between a hollow or tubular organ and the body surface.
- ~~T~~ B. Hypoplasia - underdevelopment of an organ or tissue.
- ~~T~~ C. Lobar pneumonia is most frequently (90%-95%) caused by pneumococci (*Streptococcus pneumoniae*).
- ~~F~~ D. Lobar pneumonia affects a large part, or the entirety, of a lobe and relatively common in infancy and old age. — *wrong part*
- ~~T~~ E. Cough and fever with purulent or 'rusty' sputum are common characteristics of lobar pneumonia.

2. Analyse the following statements and decide whether they are true or false:-

- ~~T~~ A. Most laryngeal carcinomas arise on the vocal cords.
- ~~F~~ B. Type I pneumocytes secrete surfactant, and replicate quickly after injury to alveolar walls.
- ~~T~~ C. Type I pneumocytes cover 95% of the alveolar surface.
- ~~F~~ D. Haemangioma and squamous papilloma are common malignant lesions of the nasal passages. *wrong (correct: uncommon)*
- ~~F~~ E. Juvenile angiofibromas are common and occur exclusively in males.

3. Analyse the following statements of tuberculosis and decide whether they are true or false:-

- ~~T~~ A. Lung is commonest site for tuberculosis.
- ~~F~~ B. Chronic alcoholism, diabetes mellitus, immunosuppression are not predisposing conditions.
- ~~T~~ C. Miliary tuberculosis may be a consequence of either primary or secondary tuberculosis.
- ~~T~~ D. A major cause of death in developing countries.
- ~~T~~ E. Most secondary tuberculosis represents reactivation of old primary infection.

4. Bronchopneumonia has a characteristic:

- ~~T~~ A. Patchy consolidation centred on bronchioles or bronchi.
- ~~T~~ B. Unusually in infancy or old age.
- ~~T~~ C. Usually secondary to pre-existing disease.
- ~~T~~ D. May occur in patients with acute bronchitis, chronic obstructive airways disease or cystic fibrosis.
- E. Pleural exudate is common

5. Squamous cell carcinomas of the lung:

- ~~F~~ A. Are more common in women than in men and are closely correlated with a smoking history.
- ~~T~~ B. Tend to arise centrally in major bronchi and eventually spread to local hilar nodes.
- ~~F~~ C. Large lesions may undergo central necrosis but central cavitation is not frequent.
- ~~T~~ D. Are often preceded for years by squamous metaplasia or dysplasia in the bronchial epithelium.
- ~~F~~ E. Squamous cell carcinomas show an excellent initial response to chemotherapy.

Miliary TB - life threatening
- occurs when large no. of bacteria travel through blood stream and spread throughout the body.

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6. Adenocarcinoma of the lungs:

- ☒ A. Adenocarcinomas are usually more peripherally located.
- ☒ B. Adenocarcinomas are the most common type of lung cancer in women and smokers.
- ☒ C. Bronchioloalveolar carcinoma is a distinctive subtype of adenocarcinoma.
- ☒ D. Adenocarcinomas are less responsive to chemotherapy.
- ☒ E. Adenocarcinomas tend to metastasize widely at an early stage

7. Atopic bronchial asthma:

- ☒ A. Most common type of asthma.
- ☒ B. A positive family history of atopy is not common.
- ☒ C. Asthmatic attacks are often preceded by allergic rhinitis.
- ☒ D. Is triggered by environmental antigens.
- ☒ E. Is triggered by some pharmacologic agents

8. Bronchiectasis:

- ☒ A. Is characterized by reversible bronchoconstriction.
 - ☒ B. Is defined as persistent productive cough for at least 3 consecutive months in at least 2 consecutive years.
 - ☒ C. Can complicate atopic asthma and chronic bronchitis.
 - ☒ D. Bronchiectatic involvement of the lungs usually affects the lower lobes bilaterally.
 - ☒ E. May develop clubbing of the fingers
9. Small (oat) cell carcinoma of the lungs:
- ☒ A. Is a highly malignant epithelial tumour of the lungs.
 - ☒ B. These tumors grow and metastasize slowly.
 - ☒ C. Small cell carcinomas are often responsible for paraneoplastic syndromes.
 - ☒ D. Small cell carcinomas are usually perihilar masses, often with extensive lymph node metastases.
 - ☒ E. Small cell carcinomas grow in various patterns, including acinar, papillary, micropapillary, and solid with mucin formation.

10. The following cells are found in chronic inflammation

- ☒ A. Macrophages
- ☒ B. Basophils
- ☒ C. Epithelioid histiocytes (stationary phagocytic cell present in CT).
- ☒ D. Lymphocytes
- ☒ E. Neutrophil

11. About necrotic granuloma

- ☒ A. Has a peripheral necrosis
- ☒ B. Mainly caused by mycobacterium tuberculosis.....
- ☒ C. Has a central necrosis.....
- ☒ D. Fungal infections can cause it.....
- ☐ E. Only occurs in the lymph nodes

Granuloma - A clump of cells that forms when the immune system tries to fight off a harmful substance but cannot remove it from the body.

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12. Which of the following statements are correct about oral herpes simplex

- ☒ A. May result into deep red painless ulcer
- ☒ B. May result into shallow red painful ulcer.....
- ☒ C. Herpes Simplex Virus 1 infections normally contracted in adulthood
- ☒ D. Lesions heal, but virus remains dormant in ganglia of trigeminal nerve.....
- ☒ E. Stress and sunlight causes reactivation of virus

13. Which of the statements are true about mucoepidermoid carcinoma

- ☒ A. Malignant tumor composed of mucinous and squamous cells.....
- ☒ B. Usually arises in parotid.....
- ☒ C. Commonly involves facial nerve Pain and numbness.....
- ☒ D. Burning or pricking sensation or loss of facial movement may be involved.....
- ☒ E. Usually arises in sublingual

14. The following statements are true about achalasia

- ☒ A. Important cause of esophageal obstruction.....
- ☒ B. Disordered esophageal motility with inability to relax LES.....
- ☒ C. Complete relaxation of lower esophageal sphincter
- ☒ D. Can be primary (idiopathic) or secondary.....
- ☒ E. Chagas's disease may cause loss of myenteric ganglion cells.....

15. Which of the following could be a site for peptic ulcer

- ☒ A. Stomach
- ☒ B. Duodenum.....
- ☒ C. Gastroesophageal junction.....
- ☒ D. Duodenum, stomach and jejunum in patients with Zollinger-Ellison syndrome.....
- ☒ E. Meckel diverticulum with ectopic gastric mucosa.....

16. The following cases require a mandatory forensic post mortem examination to ascertain the cause of death

- ☒ A. Road traffic accident
- ☒ B. Suicide
- ☒ C. Sudden cardiac death
- ☒ D. Death in hospital from pneumonia 6 months after gun shot
- ☒ E. Snake bite

17. An abnormal variation in structure and function of any part of the body is

- ☒ A. Pathogenesis
- ☒ B. Aberration
- ☒ C. Pathology
- ☒ D. Disease
- ☒ E. Diagnosis

18. Characteristic features of anaplasia include the following:
- ☒ A. Prominence of the nucleoli
 - ☒ B. Loss of orientation of cells
 - ☒ C. Loss of architecture
 - ☒ D. Atypical mitoses
 - ☒ E. Increased cytoplasmic to nucleus ratio
19. Liquefactive necrosis occurs in the following settings:
- ☒ A. In bacterial infection
 - ☒ B. In venous obstruction
 - ☒ C. In arterial obstruction by a thrombus
 - ☒ D. When there is release of pancreatic lipase
 - ☒ E. In gangrene of limb
20. Which infectious diseases cause vasculitis?
- ☒ A. Viral infection
 - ☒ B. Bacterial sepsis
 - ☒ C. Syphilis
 - ☒ D. Fungi
 - ☒ E. Parasitic infection
21. Where are most syphilitic aneurysms located?
- ☒ A. Ascending aorta
 - ☒ B. Abdominal aorta
 - ☒ C. Arch of the aorta
 - ☒ D. Aortic bifurcation
 - ☒ E. Iliac arteries
22. Unconjugated hyperbilirubinemia is associated with
- ☒ A. Raised unconjugated bilirubin in blood x None in urine
 - ☒ B. Raised conjugated bilirubin in bile - dark
 - ☒ C. Raised urobilinogen in urine
 - ☒ D. Raised conjugated bilirubin in blood + in urine - dark
 - ☒ E. Raised alkaline phosphatase
23. About cholecystitis:
- ☒ A. Acute: is associated with fever, leukocytosis, or right upper quadrant pain
 - ☒ B. Chronic: presents with subclinical pain
 - ☒ C. Ultrasound can detect stones well
 - ☒ D. Go hand in hand with stones in gallbladder or ducts, age, sex, weight
 - ☒ E. If surgery is required, most is laparoscopic
24. Edema:
- ☒ A. Signifies increased blood in the interstitial tissue spaces
 - ☒ B. Involves decreased vascular permeability
 - ☒ C. Is an adaptive response
 - ☒ D. Is the accumulation of fluid in the interstitial spaces
 - ☒ E. As a characteristic feature of chronic inflammation

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25. Which of the following is NOT a feature of the Virchow's triad?
☐ A. Endothelial injury
☐ B. Hypercoagulability
☐ C. Abnormal blood flow
☒ D. Angiogenesis
☐ E. Turbulent flow of blood
26. The following cells are found in chronic inflammation
☐ A. Macrophages
☐ B. Basophils
☒ C. Epithelioid histiocytes
☐ D. Lymphocytes
☐ E. Neutrophils
27. About necrotic granuloma
☐ A. Has a peripheral necrosis
☐ B. Mainly caused by mycobacterium tuberculosis.
☐ C. Has a central necrosis
☐ D. Fungal infections can cause it
☐ E. Only occurs in the lymph nodes
28. About Healing and repair
☐ A. Initiated after acute inflammation
☐ B. Initiated when inflammation begins
☐ C. Regeneration is the key which is by production of fibrous material
☐ D. Done via combination of regeneration and repair
☒ E. Occurs in the same way in all types of cells
29. Examples of labile cells are
☒ A. Hepatocytes
☐ B. Cells in the crypt of intestines
☐ C. Skin cells
☐ D. Cardiac Muscle cells
☐ E. Nerve cells
30. About oxygen dependent killing pathway of the phagocytes
☒ A. This is the less effective way of killing the pathogen by a phagocytes
☐ B. Results into HOCL generated by oxidative enzymes
☐ C. H_2O_2 converted to HOCL by SOD
☐ D. O^2 converted to H_2O_2 by MPO
☒ E. O^2 converted to H_2O_2 by SOD

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SECTION C. MATCHING QUESTIONS

GIVE TO ITEMS IN THE RIGHT COLUMN A LETTER ACCORDING TO THEIR MATCH IN THE LEFT COLUMN (CHOOSE THE MOST APPROPRIATE ITEM). FOR ANSWER USE COLUMN '1'.

1.

1	2	3	4
E	Klebsiella pneumoniae	A.	Caused by organization of exudate and persistence of infection.
A	Chronic pneumonia	B.	Are prone to rupture, causing spontaneous pneumothorax.
B	Bullous emphysema	C.	Intrapleural fibrinosuppurative reaction
		D.	The bronchioles and the adjacent alveoli are filled with exudate consisting chiefly of neutrophils.
		E.	Thick and gelatinous sputum

#2.

1	2	3	4
E	Juvenile angiofibroma	A	Loss of pulmonary surfactant
A	Patchy atelectasis	B	Overgrowth of mature tissue in which the elements show disordered arrangement and proportion in comparison to normal.
C	Red hepatization.	C	Lasts for a few days; massive accumulation in the alveolar spaces of polymorphs, together with some lymphocytes and macrophages. Many red cells are extravasated.
		D.	The bronchioles and the adjacent alveoli are filled with exudate consisting chiefly of neutrophils.
		E.	Extremely vascular and surgical removal can be difficult; contain androgen receptors

#3

1	2	3	4
C	Empyema	A.	It results from the lytic action of neutrophils and is most often found in pneumonia caused by Staphylococcus aureus.
A	Abscess	B.	are prone to rupture, causing spontaneous pneumothorax.
B	Bronchopneumonia	C.	Intrapleural fibrinosuppurative reaction
		D.	The bronchioles and the adjacent alveoli are filled with exudate consisting chiefly of neutrophils.
		E.	Premature infants

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#4

1	2	3	4
B	Nasal polyps	A.	May cause metastatic abscesses, endocarditis, meningitis, or suppurative arthritis.
P	Bacteremia with hematogenous dissemination	B.	Polypoid oedematous masses of connective tissue infiltrated with chronic inflammatory cells.
D	6. Hyaline membrane disease	C.	Overgrowth of mature tissue in which the elements show disordered arrangement and proportion in comparison to normal.
		D.	Premature infants
		E.	Anaplastic undifferentiated cells growing without any distinct pattern.

#5

1	2	3	4
E	Oat cell carcinoma	A.	Adenocarcinoma
D	Pancoast syndrome	B.	Polypoid oedematous masses of connective tissue infiltrated with chronic inflammatory cells.
A	The most common type of lung cancer in women	C.	Pneumonia, abscess, and lobar collapse
		D.	Depression of the eyeball. Ptosis of the upper eyelid. Constriction of the pupil. Absence of sweating.
		E.	Tumor consists of small oval to spindle-shaped cells with scant cytoplasm, finely granular nuclear chromatin and conspicuous mitoses.

SECTION D. SHORT ESSAYS. 10 MARKS EACH

PART A. CHOOSE 2 ESSAYS.

1. a. Describe two types of atresia .
b. Briefly, discuss their clinical signs
2. Briefly, discuss Pleomorphic Adenoma .
3. Discuss autoimmune gastritis

PART B. CHOOSE 3 ESSAYS.

1. Discuss etiology, pathogenesis and morphology of lobar pneumonia
2. Discuss etiology and morphology of Carcinoma of the Lung.
3. Discuss Pneumoconioses.
4. Discuss Obstructive Pulmonary Diseases.
5. Tuberculosis Is the Classic Granulomatous Infection. Discuss Tuberculosis.

PART 3. CHOOSE 2 ESSAYS.

- 1 Discuss the morphological patterns of necrosis. ✓
- 2 Briefly highlight the complications of myocardial infarction x
- 3 Write briefly about the types of cellular adaptation ✓